

1. Algorithm

Step 1: Start

Step 2: Declare variables choice, i, n, n1, n2, key, found, and required arrays arr[], arr1[], arr2[], merged[].

Step 3: Display the menu:

1. Count even, odd, positive, negative, and zero elements
2. Search for an element using linear search
3. Merge 2 arrays
4. Exit

Step 4: Read choice from the user.

Step 5: Perform operation based on choice:

- **If choice == 1:**

1. Input size of array n.
2. Input n elements in array arr[].
3. Initialize even = 0, odd = 0, pos = 0, neg = 0, zero = 0.
4. Loop i from 0 to n-1:
 - If arr[i] > 0, pos++.
 - Else if arr[i] < 0, neg++.
 - Else zero++.
 - If arr[i] % 2 == 0, even++ else odd++.
5. Print all counts.

- **If choice == 2:**

1. Input size of array n.
2. Input n elements in array arr[].
3. Input key element to search.
4. Set found = 0.
5. Loop i from 0 to n-1:
 - If arr[i] == key, print element found at index i, set found = 1, and break loop.
6. If found == 0, print element not found.

- **If choice == 3:**

1. Input size of first array n1.
2. Input n1 elements in arr1[].
3. Input size of second array n2.
4. Input n2 elements in arr2[].

5. Loop i from 0 to $n1-1$: $merged[i] = arr1[i]$.
6. Loop i from 0 to $n2-1$: $merged[n1 + i] = arr2[i]$.
7. Print merged array.

- **If choice == 4:**

1. Exit program.

- **Else:**

1. Print invalid choice.

Step 6: Repeat steps 3–5 using a loop until choice is 4.

Step 7: Stop

2. Source Code

```
#include <stdio.h>

int main() {
    int choice;

    do {
        printf("
--- MENU ---
");
        printf("1. Count Even, Odd, Positive, Negative, and Zero elements
");
        printf("2. Search element using Linear Search
");
        printf("3. Merge 2 Arrays
");
        printf("4. Exit
");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1: {
                int n, i;
                int even = 0, odd = 0, pos = 0, neg = 0, zero = 0;

                printf("Enter the number of elements: ");
                scanf("%d", &n);

                int arr[n];
                printf("Enter %d elements: ", n);
                for (i = 0; i < n; i++) {
                    scanf("%d", &arr[i]);
                }

                for (i = 0; i < n; i++) {
                    if (arr[i] > 0) {
                        pos++;
                    }
                }
            }
        }
    } while (choice != 4);
}
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        } else if (arr[i] < 0) {
            neg++;
        } else {
            zero++;
        }

        if (arr[i] % 2 == 0) {
            even++;
        } else {
            odd++;
        }
    }

    printf("Positive numbers: %d
", pos);
    printf("Negative numbers: %d
", neg);
    printf("Zeroes: %d
", zero);
    printf("Even numbers: %d
", even);
    printf("Odd numbers: %d
", odd);
    break;
}

case 2: {
    int n, i, key, found = 0;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    int arr[n];
    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    printf("Enter element to search: ");
    scanf("%d", &key);

    for (i = 0; i < n; i++) {
        if (arr[i] == key) {
            printf("Element %d found at index %d
", key, i);

            found = 1;
            break;
        }
    }

    if (found == 0) {
        printf("Element %d not found in the array
", key);
    }
    break;
}

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    }

    case 3: {
        int n1, n2, i;

        printf("Enter size of first array: ");
        scanf("%d", &n1);
        int arr1[n1];
        printf("Enter elements of first array: ");
        for (i = 0; i < n1; i++) {
            scanf("%d", &arr1[i]);
        }

        printf("Enter size of second array: ");
        scanf("%d", &n2);
        int arr2[n2];
        printf("Enter elements of second array: ");
        for (i = 0; i < n2; i++) {
            scanf("%d", &arr2[i]);
        }

        int merged[n1 + n2];

        for (i = 0; i < n1; i++) {
            merged[i] = arr1[i];
        }
        for (i = 0; i < n2; i++) {
            merged[n1 + i] = arr2[i];
        }

        printf("Merged Array: ");
        for (i = 0; i < n1 + n2; i++) {
            printf("%d ", merged[i]);
        }
        printf("
");
        break;
    }

    case 4:
        printf("Exiting program...
");
        break;

    default:
        printf("Invalid choice! Please try again.
");
    }
} while (choice != 4);

return 0;
}

```

3. Test Case Table

Operation / Option	Input Data	Expected Output	Actual Result	Status
Option 1: Count Elements	Size: 5 Array: [12, -5, 0, 7, -2]	Positive: 2 Negative: 2 Zero: 1 Even: 3 Odd: 2	Positive: 2 Negative: 2 Zero: 1 Even: 3 Odd: 2	Pass
Option 2: Linear Search (Found)	Size: 4 Array: [10, 20, 30, 40] Search Key: 30	Element 30 found at index 2	Element 30 found at index 2	Pass
Option 2: Linear Search (Not Found)	Size: 3 Array: [1, 2, 3] Search Key: 5	Element 5 not found in the array	Element 5 not found in the array	Pass
Option 3: Merge Arrays	Array 1: [1, 2] Array 2: [3, 4, 5]	Merged Array: 1 2 3 4 5	Merged Array: 1 2 3 4 5	Pass
Option 4: Exit	Choice: 4	Exiting program...	Exiting program...	Pass